Part E · Justification of the clinical review figure

Name: _____ Roll no: _____ Data: vitals_10k.csv, synthetic, CDB601220.

1. Per panel: question, chart, channel rule

Panel 1 (systolic histogram). *Is systolic pressure one population?* A histogram shows a whole distribution, which a mean or box would hide: here two humps with a valley at 141 mmHg, 42% of patients in the high one. Rule followed: **position along a common scale** (bar tops on one baseline) for the quantity, colour only for the two groups, stacked so the total shape stays visible.

Panel 2 (length of stay by ward). *Which wards keep patients longest?* Box plots compare medians and spread across four categories where means are wrecked by skew. Wards are sorted by median so rank is read from position; only ICU is highlighted, the rest muted, so one colour carries one message.

Panel 3 (age vs cholesterol scatter). *Is the one correlated pair a real linear trend?* Two continuous variables call for position on two aligned axes, the most accurate channel. Panel 4 chose the pair; the fitted line (0.21 mmol/L per decade) is drawn on all 10,000 rows.

Panel 4 (correlation heatmap). *Which pairs are worth a scatter?* A matrix is a screening tool for 36 pairs. Rule: a **diverging** palette (RdBu_r) because correlation has a sign, centred at 0 with vmin/vmax = -1/+1 so colour is comparable, upper triangle masked, every cell annotated so colour is never the only carrier of the number.

2. Every parameter, defended

binwidth = 2 mmHg (panel 1): tried 1, 2, 5, 10. 1 mmHg gives 6 false peaks on whole-number data; 5 and 10 hide the shape of the valley; 2 is the narrowest width that keeps two clean humps. **Valley = 141 mmHg** is computed (minimum of a 5-bin moving average between 130 and 160), not eyeballed.

Log y-axis (panel 2): length of stay spans 0.01 to 30 days and is right-skewed, so a linear box collapses to a line. Consequence: the 2 stays of exactly 0 days cannot be drawn on a log axis and are not shown. **No fliers hidden** (showfliers left on).

Scatter: subsample 2,000 of 10,000 (seed 7) and alpha 0.35 to avoid a solid blob; the line is fitted on all rows. **Heatmap:** cmap RdBu_r, vmin -1, vmax 1, center 0, fmt .2f, upper triangle masked; map_mmhg left out. **Palette:** Okabe-Ito blue/orange from vizlib, colour-blind safe (worst pair under simulation: blue/green under tritan, dE 8.4, still above the 8 target). **Export:** 300 dpi, tight bbox, source note on the figure.

Cleaning rules (thresholds chosen from physiology and sentinel values before looking at any effect): heart rate outside 30 to 250 bpm; BMI of 0 or 999; length of stay over 60 days.

3. Every row removed: rule, count, before and after

Wards: 12 spellings conformed to 4 (strip, lower-case, map). 0 rows removed. **Dates:** 3 formats parsed explicitly, slash dates day-first (1,933 have a first field above 12 and none a second field above 12). 0 rows removed.

BMI sentinels (0 or 999 meaning missing): n = 486 (402 at 999, 84 at 0), set to missing. Mean BMI moves from **66.18** to **27.35** kg/m².

Length of stay above 60 days: n = 18 (140 to 274 days, which is 6 to 11 days if they were typed in hours; I excluded rather than converted because that is a guess). Mean moves from **2.31** to **1.95** days.

map_mmhg is computed as (systolic + 2 x diastolic)/3, so its correlation with both is arithmetic; excluded from the heatmap, no rows lost.

Heart rate (the ethics clause). Readings above 250 bpm were removed as device faults (n = 29), as were 8 readings of 0 bpm (below 30), 37 values in total, with the rest of each row kept. Mean heart rate moves from **78.83** to **78.05** bpm. The figure was also produced without the exclusion; it is in the appendix (partD_appendix_heart_rate.png). The exclusion moves no correlation with heart rate by more than 0.02, and heart rate is in none of the headline claims.

Symmetry test. I would have made the same removal if it weakened my conclusion: the 30 to 250 window is physiological, was fixed before I computed any effect, and no human heart beats at 0 or 400 bpm whichever way the chart then looks. Also not removed but declared: 304 SpO2 readings sit at the 100% ceiling (kept) and risk_score is a ratio and is never averaged across wards.